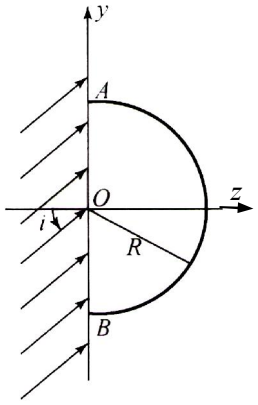
$e^2/2C_\Sigma$; use C_GV_G horizontally, with C_GV_G in units of e. Mark key points. For $n=0$, 1, 2, 3, enter C_GV_G/e ranges below. (Only the graph and Table 1 are scored.)

Table 1

n	0	1	2	3
C_GV_G/e range				

5.

A clear semicylinder, index $n=1.50$, radius R, is in air. The figure is a normal section; O is its axis intersection. Light enters only through plane AB. A broad parallel monochromatic beam, normal to the axis, covers AB at angle i. Some light passes through and exits the curved surface. Its width on AB along y is d. Exclude rays reflected one or more times inside the cylinder before exiting.



1. As incidence angle i varies over $0°\sim90°$, find the minimum d_{min} of d, and its maximum d_{max}.

2. In the figure plane, relate the arc subtended at O by the exit beam to i. Also locate this arc at grazing incidence.

6.

General relativity bends light in a stellar gravity field. The ray is in a plane through the centre, bending slightly near it. Its closest point is periapsis. A line through periapsis and centre is its symmetry axis. In polar coordinates (r, φ), take the centre as origin, symmetry axis as polar axis. The photon trajectory is

$$r=\frac{GM/c^2}{a\cos\varphi+a^2(1+\sin^2\varphi)},$$

G is the gravitational constant, M stellar mass, c light speed; a is far below 1 in magnitude. A star is 80.0 light years away; a white dwarf lies halfway to Earth. At Earth, rays past either side subtend 1.80×10^{-7} rad. Find the dwarf mass (kg).

Given $G=6.6730\times10^{-11}\ m^3/(kg\cdot s^2)$